



Principles of Simulation

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Computational Simulation

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Outline



- Definition of Simulation
- Modeling Concepts
- Modeling Structures
- Advantages and Disadvantages of Simulation
- Areas of Application
- Steps in a Simulation Study
- Random Number and Random Variate Generation
- Input Data
- Verification and Validation
- Experimentation and Output Analysis

Definition of Simulation

Definition of Simulation



Simulation is the **imitation** of the operation of a **real-world process** or **system** over **time**.

Simulation involves the **generation** of an **artificial history** of the **system** and the observation of that artificial history **to draw inferences** concerning the **operating characteristics** of the real system that is represented.

Simulation is an **indispensable problem-solving methodology** for the solution of many real-world problems.

Simulation is used to **describe and analyze** the **behavior of a system**, **ask what-if questions** about the real system, and aid in the design of real systems.

Both **existing and conceptual systems** can be modeled with simulation.

Example Ad Hoc Simulation



Consider the operation of a **one-teller bank** where **customers arrive for service between 1 and 10 minutes apart in time**, integer values only, each value equally likely. The customers are **served in between 1 and 6 minutes**, also integer valued, and equally likely. Restricting the times to integer values is an abstraction of reality since time is continuous, but this aids in presenting the example.

The objective is to simulate the bank operation, by hand, **until 20 customers are served**, and to compute measures of performance such as the percentage of idle time, the average waiting time per customer, and so on. Admittedly, 20 customers are far too few to draw conclusions about the operation of the system for the long run, but by following this example, the stage is set for further discussion in this chapter and subsequent discussion about using the computer for performing simulation.

Example Ad Hoc Simulation



To simulate the process, random interarrival and service times need to be generated. Assume that the interarrival times are generated using a spinner that has possibilities for the values 1 through 10. Further assume that the service times are generated using a dice that has possibilities for the values 1 through 6.

Example Ad Hoc Simulation

Spinner

<https://www.google.com/search?q=spinner+wheel>
<https://www.visnos.com/demos/random-spinners>



Die

<https://www.google.com/search?q=Roll%20a%20die>
<https://www.online-stopwatch.com/online-dice/>



(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Customer	Time Between Arrivals	Arrival Time	Service Time	Service Begins	Time Service Ends	Time in System	Idle Time	Time in Queue
1	-	0	2	0	2	2	0	0
2	5	5	2	5	7	2	3	0
3	1	6	6	7	13	7	0	1
4	10	16	5	16	21	5	3	0
5	6	22	6	22	28	6	1	0
6	2	24	4	28	32	8	0	4
7	9	33	3	33	36	3	1	0
8	1	34	4	36	40	6	0	2
9	10	44	1	44	45	1	4	0
10	3	47	3	47	50	3	2	0
11	5	52	1	52	53	1	2	0
12	2	54	2	54	56	2	1	0
13	3	57	3	57	60	3	1	0
14	5	62	6	62	68	6	2	0
15	4	66	2	68	70	4	0	2
16	3	69	6	70	76	7	0	1
17	7	76	4	76	80	4	0	0
18	8	84	5	84	89	5	4	0
19	7	91	3	91	94	3	2	0
20	7	98	1	98	99	1	4	0
						79	30	10

Example Ad Hoc Simulation - Performance measures



Average time in system = $79/20 = 3.95$ minutes.

Percent idle time = $(30/99)(100) = 30\%$.

Average waiting time per customer = $10/20 = 0.5$ minute.

Fraction having to wait = $5/20 = 0.25$.

Average waiting time of those who waited = $10/5 = 2$ minutes.

Conclusions



- This small simulation can be accomplished by hand, but there is a limit to the complexity of problems that can be solved in this manner.
- The number of customers that must be simulated could be much larger than 20.
- The number of times that the simulation must be run for statistical purposes could be large.
- Hence, using the computer to solve real simulation problems is almost always appropriate.

Issues



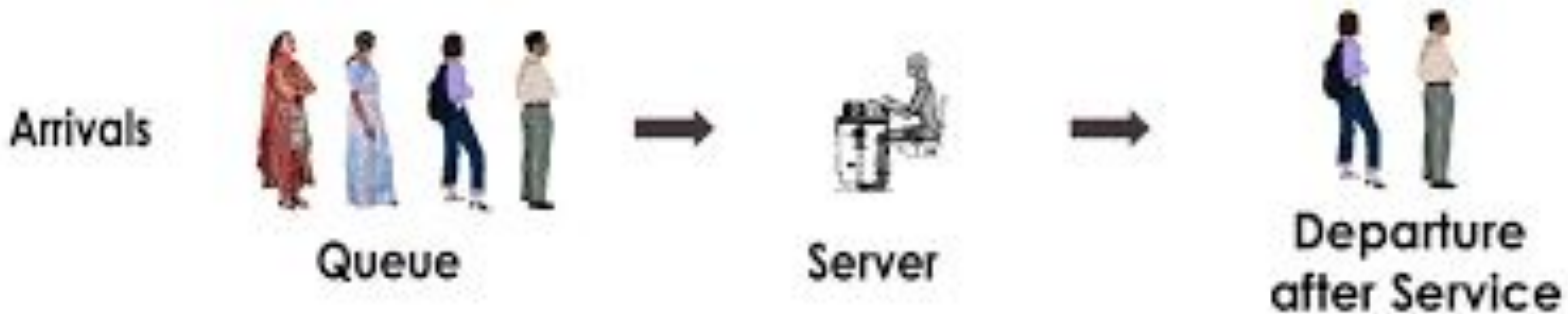
1. How is the form of the input data determined?
2. How are random variates generated if they follow statistical distributions other than the discrete uniform?
3. How does the user know that the simulation imitates reality?
4. What other kinds of problems can be solved by simulation?
5. How long does the simulation need to run?
6. How many different simulation runs should be conducted?
7. What statistical techniques should be used to analyze the output?

Modeling Concepts

System, Model, and Events

- A **model** is a representation of an actual **system**. (complex but not too much)
- Consider an **event** as an occurrence that changes the state of the system (exogenous events and endogenous events).

Single Server queue model



System, Model, and Events



There are:

- Discrete-event simulation models
- Descriptive models
- Mathematical models
- Statistical models
- Input-output models

System, Model, and Events



There are:

- Discrete-event simulation models
- Descriptive models
- Mathematical models
- Statistical models
- Input-output models



A discrete-event model attempts to represent the components of a system and their interactions to such an extent that the objectives of the study are met.
(DES is *dynamic*)

Most of them represent a system's inputs and outputs explicitly but represent the internal of the model with mathematical or statistical relationships. Most of them are *static*.

An example of mathematical model from physics based on theory:

$$\text{force} = \text{mass} \times \text{acceleration}$$

System State Variables



- These are the collection of all information needed to define what is happening within a system to a sufficient level (i.e. to attain the desired output) at a given point in time.
- Determining the system state variables is as much an art as a science.
- During modeling process, any omissions will readily come to light. (On the other hand, unnecessary state variables may be eliminated).
- The system state variables in a *discrete-event model* remain constant over intervals of time and change value only at certain well-defined points called ***event times***.
- *Continuous models* have system state variables defined by differential or difference equations, giving rise to variables that may change continuously over time.

Entities and Attributes



- An **entity** represents an object that requires explicit definition.
 - a. It can be **dynamic** (moves through the system) or **static** (serve other entities).
- An entity may have **attributes** that pertain to that entity alone.
 - a. They should be considered as local values.
 - b. Example: **time of arrival**

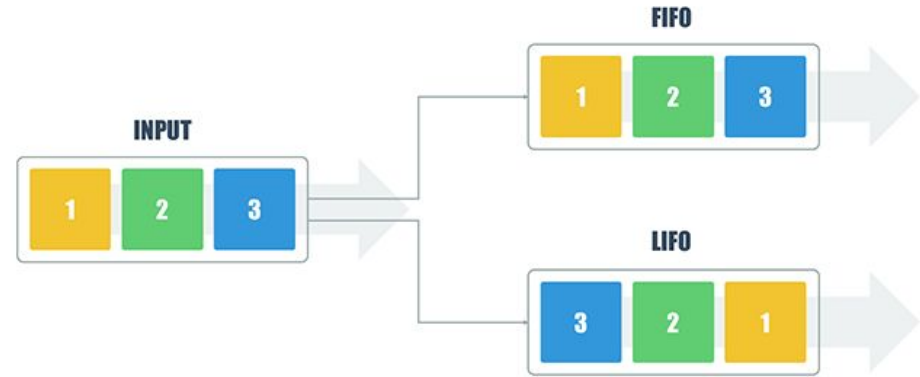
Resources



- A **resource** is an entity that provides service to dynamic entities.
- The resource can serve one or more than one dynamic entity at the same time (i.e., operate as a parallel server).
- A dynamic entity can request one or more units of a resource.
 - a. If denied, the requesting entity **joins a queue** or takes some other action (i.e., is diverted to another resource, is ejected from the system). (Other terms for queues are *files*, *chains*, *buffers*, and *waiting lines*).
 - b. If permitted to capture the resource, the entity remains for a time, then releases the resource.
 - i. There are many possible states of a resource.
 - ii. Minimally, these states are **idle** and **busy**. But other possibilities exist, including **failed**, **blocked**, or **starved**.

List Processing

- Entities are managed by allocating them to resources that provide service; by attaching them to event notices, thereby suspending their activity into the future; or by placing them into an ordered list.
- Lists are used to represent **queues**.
- Lists are often processed according to:
 - a. **FIFO** (first in, first out)
 - b. **LIFO** (last in, first out),
 - c. **Randomly**
 - d. **According to the value of an attribute**



E.g. SPT (shortest process time) scheduling. In this case the processing time may be stored as an attribute of each entity. The entities are ordered according to the value of that attribute, with the lowest value at the head or front of the queue.

Activities and Delays



- An **activity** is a period of time whose duration is known prior to commencement of the activity. Thus, when the duration begins, its end can be scheduled.
 - a. The duration can be a constant (e.g., 10 minutes)
 - b. Random value from a statistical distribution (e.g., exponential distribution with a mean 10 min)
 - c. Result of an equation: e.g., $duration = \begin{cases} 0.9 \times value & , 0 < clocktime \leq 4 \\ 1.1 \times value & , 4 < clocktime \end{cases}$
 - d. Input from a file
 - e. Computed based on the event state (e.g., 10 minutes when the preceding queue contains at most four entities and 8 minutes when there are five or more in the preceding queue).
- A **delay** is an indefinite duration that is caused by some combination of system conditions.

Discrete-Event Simulation (DES) Model



- Discrete-Event Simulation can be defined as one in which the state variables change only at those discrete points in time at which events occur.
- Events occur as a consequence of activity time and delays.
- Entities may compete for system resources, possibly joining queues while waiting for an available resource.
- Activity and delay times may “hold” entities for durations of time.
- A DES model is conducted over time (“run”) by a mechanism that moves simulated time forward.
- The system state is updated at each event, along with capturing and freeing of resources that may occur at that time.

Modeling Structures

Modeling Structures



- Process-Interaction Method
- Event-Scheduling Method
- Activity Scanning
- Three-Phase Method

Process-Interaction Method

- The notion is that the computer program should **emulate** the **flow** of an object through the system.
- The **entity moves** as far as possible in the **system** until it is **delayed**, **enters** an activity, or **exits** from the system.
- When the **entity's movement is halted**, the **clock advances** to the **time of the next movement of any entity**.
- This **flow**, or movement, **describes** in **sequence all the states** that the object can attain in the system.



Event-Scheduling Method

- The basic concept is to advance time to when something next happens.
- This usually **releases** a **resource** (i.e. a scarce entity such as a machine or transporter).
- The event then **reallocates** available objects or **entities** by **scheduling activities** where they can now participate.



Activity Scanning



- This is also known as the **two-phase approach**.
- Activity scanning is **similar to rule-based programming**.
 - If a specified condition is met, a rule is ***fired***, meaning that an action is taken.
- Activity scanning produces a simulation program **composed of independent modules waiting to be executed**.
- Scanning **takes place at fixed time increments** at which a **determination is made** concerning **whether or not an event occurs** at that time.
- If an event occurs, the system state is updated.

Three-Phase Method



- Time is advanced until there is a state change in the system or until something next happens.
- The system is examined to **determine all of the events that take place at this time** (i.e., all the activity completions that occur).
- Only when all resources that are due to be released at this time have been **released is reallocation of these resources into new activities** started in the third phase of the simulation.
- In summary:
 - First phase is time advance
 - The second phase is the release of those resources scheduled to end their activities at this time.
 - The third phase is to start activities given the global picture about resource availability.

Advantages and Disadvantages of Simulation

Advantages



1. Choose correctly
2. Compress and expand time
3. Understand why
4. Explore possibilities
5. Diagnose problems
6. Identify constraints
7. Develop understanding
8. Visualize the plan
9. Build consensus
10. Prepare for change
11. Invest wisely
12. Train the team
13. Specify requirements

Disadvantages



1. Model building requires special training
2. Simulation results may be difficult to interpret
3. Simulation modeling and analysis can be time consuming and expensive
4. Simulation may be used inappropriately

However...



In defense of simulation, these four disadvantages, respectively, can be offset as follows:

1. Simulators
2. Output analysis
3. Faster and faster
4. Limitations of closed-form models

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Areas of Application

Manufacturing and material Handling Applications

- Minimizing synchronization delays of prefabricated parts before assembly
- Evaluation of AGV routing strategies
- Flexible modeling and analysis of large-scale AS/RS-AGV systems
- Design and analysis of large-scale material handling systems
- Material flow analysis of automotive assembly plants
- Analysis of the effects of work-in-process levels on customer satisfaction
- Assessing the cost of quality

Solar panel production line

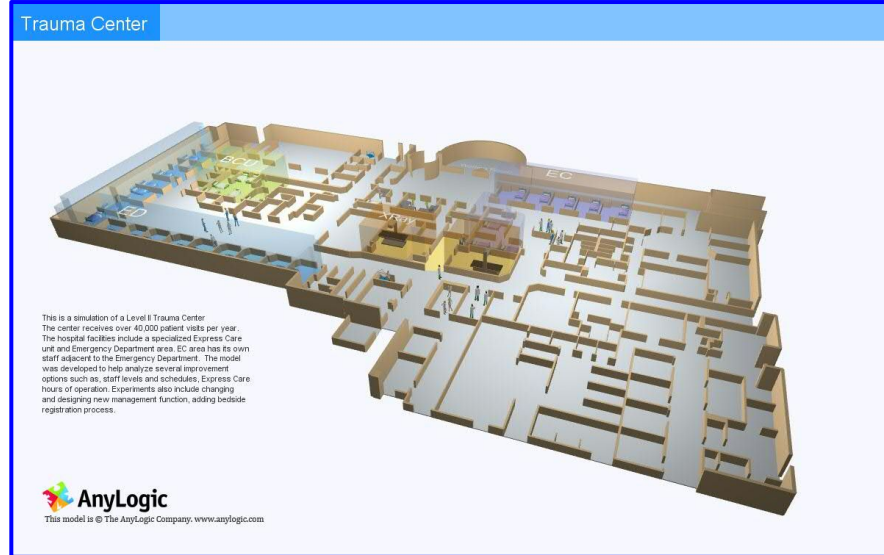


Public Systems Applications

Health Systems

- Screening for abdominal aortic aneurysms
- Lymphocyte development in immuno-compromised patients
- Asthma dynamics and medical amelioration
- Timing of liver transplants
- Diabetic retinopathy
- Evaluation of nurse-staffing and patient-population scenarios
- Evaluation of automated equipment for a clinical processing laboratory
- Evaluation of hospital surgical suite and critical care area

Trauma Center NUR 503 - Simulation Models in AnyLogic Cloud



Public Systems Applications

Military Systems

- Air Force support equipment use
- Analysis of material handling equipment for prepositioning ships
- Development and implementation of measures of effectiveness
- Reengineering traditional stovepiped Army staffs for information operations
- Evaluation of theater airlift system productivity
- Evaluation of C-141 depot maintenance
- Evaluation of air mobility command channel cargo system

Air Defense System



Border Checkpoint

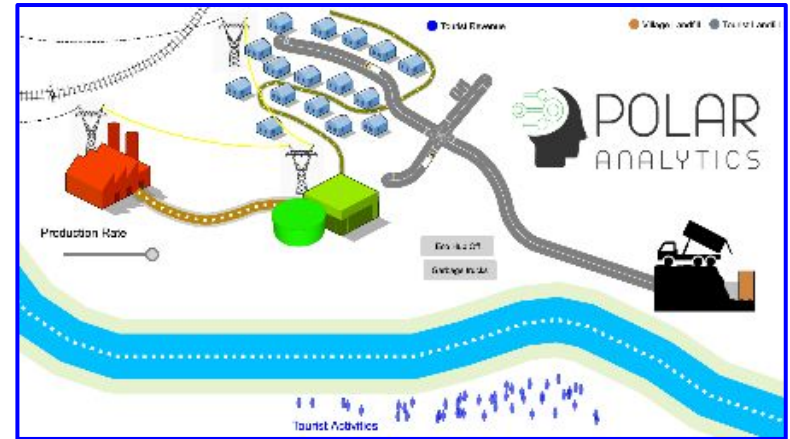


Public Systems Applications

Natural Resources

- Nonpoint-source pollution analysis
- Weed scouting and weed control decision making
- Evaluation of surface water quality data

Circular Economy Concept

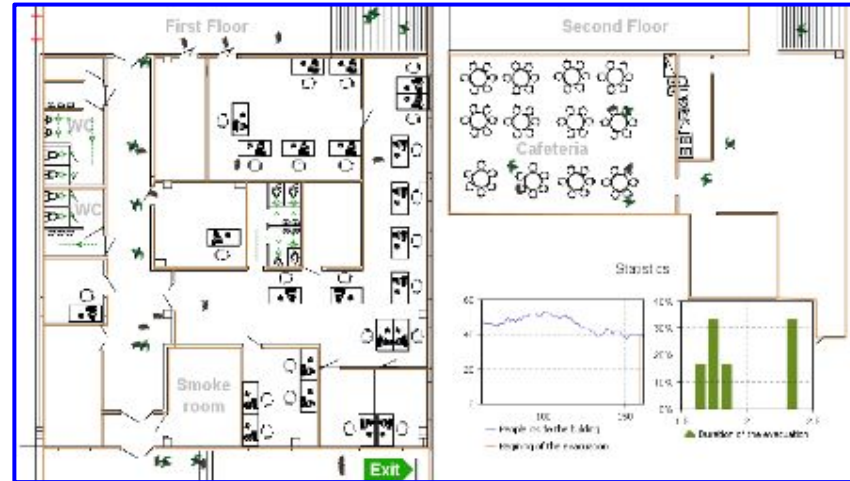


Public Systems Applications

Public Services

- Emergency ambulance system analysis
- Evaluation of flow of civil lawsuits
- Evaluation of field offices within a government agency

Evacuation



Service System Application

Transportation

- Analysis of intelligent vehicle highway systems
- Evaluation of traffic control procedures at highway work zones
- Evaluation of taxi management and route control
- Animation of a toll plaza
- Port traffic planning model analysis
- Evaluation of rapid transit modeling with automatic and manual controls

Roundabout

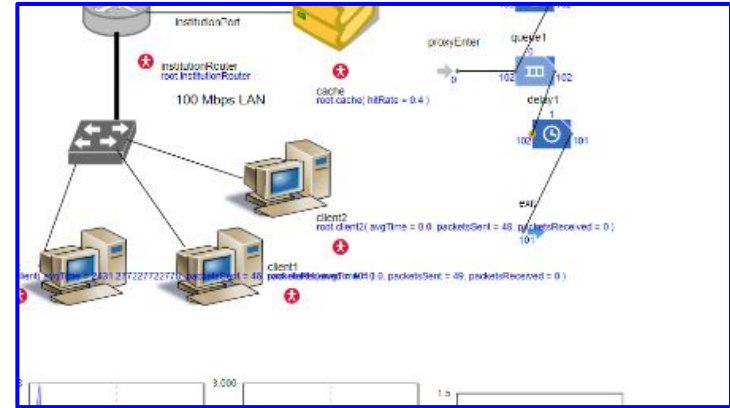


Service System Application

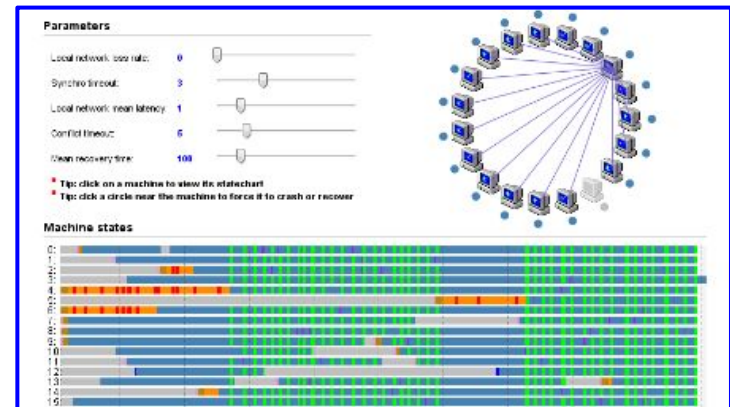
Computer Systems Performance

- User transaction processing behavior analysis
- Evaluation of database transaction management protocols
- Evaluation of analytic models of memory queueing

Proxy



Leader Election Protocol



Service System Application

Air Transportation

- Evaluation of human behavior in aircraft evacuations
- Analysis of airport/airline operations
- Evaluation of combinations carrier air cargo hub

Airport



Airplane Boarding Policy Simulator



Service System Application

Communications Systems

- Trunked radio network analysis
- Evaluation of telephone service provisioning process
- Picture archiving and communications system analysis
- Evaluation of modeling of broadband telecommunication networks
- Analysis of virtual reality for telecommunication networks

Cell Phone Users and Network Capacity



Cell Phone Users and Network Capacity

A cell telecom company is about to introduce a new value added service to its users, say, hi-speed internet access and is planning the network capacity.

Initially, all users are voice-only users. They may adopt hi-speed internet because of advertizing or because of word of mouth. Once they start using internet, they may be happy, or may get frustrated with long response times. While users are happy, they share their positive experience with others. If they are frustrated, they quit internet usage and begin to spread negative word of mouth, which may prevent voice-only users from adoption. But after a while frustrated users forget and forgive, and may consider using the service again.

The network is modeled as a simple multi-server system. Internet users periodically go online, generate transactions, and assess response times.

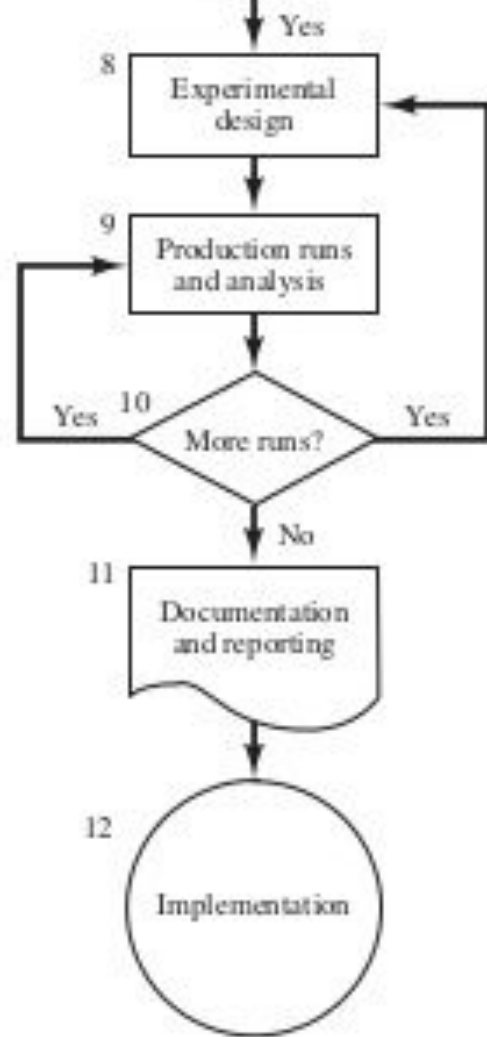
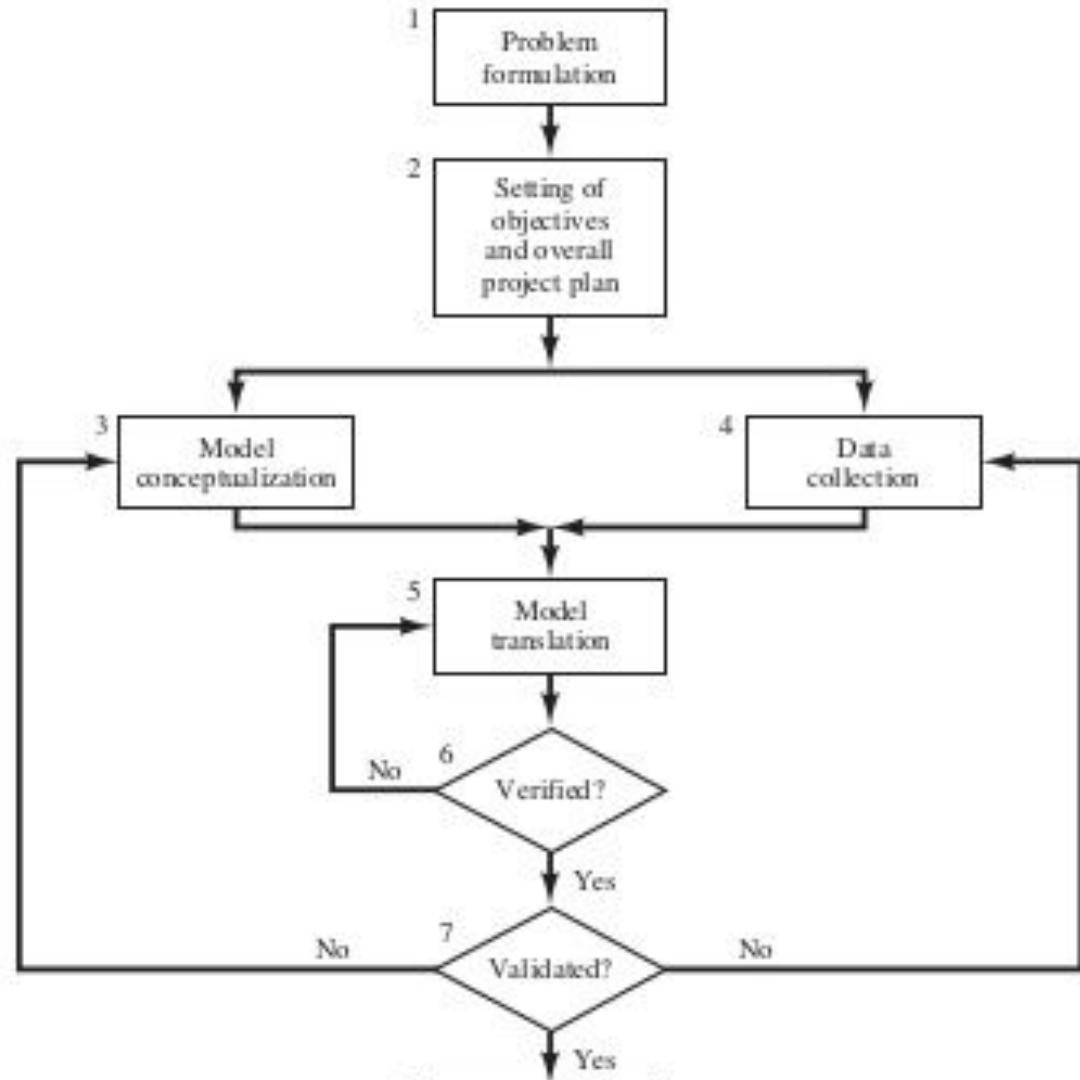
This is a multi-method model: the users are agents, and the network is modeled in discrete event style.

AnyLogic LEGACY NOTICE

Since version 7 AnyLogic offers the Process Modeling Library - a new highly improved solution for discrete event (process) modeling. The old Enterprise Library is no more included in the product.

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Steps in a Simulation Study



HOW TO CONDUCT A SUCCESSFUL SIMULATION STUDY

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ABSTRACT

In this tutorial we give a definitive and comprehensive seven-step approach for conducting a successful simulation study. Topics to be discussed include problem formulation, collection and analysis of data, developing a valid and credible model, modeling sources of system randomness, design and analysis of simulation experiments, and project management.

1 INTRODUCTION

A simulation study is a sophisticated systems-analysis activity that requires an analyst to have, at a minimum, knowledge of simulation methodology (model validation, selecting input probability distributions, design and analysis of simulation experiments, etc.), probability theory, statistics, project management, and the detailed operations of the system being studied. Model "programming" represents only 25 to 50 percent of the work in a sound simulation study, despite the fact that many organizations view simulation as little

model (model assumptions) has been correctly translated into a computer "program," i.e., debugging the simulation computer program. Although verification is simple in concept, debugging a large-scale simulation program is a difficult and arduous task due to the potentially large number of program paths. Techniques for debugging simulation programs include a structured-walkthrough of the program, use of a trace or an interactive debugger, and animation.

Validation is the *process* of determining whether a simulation model is an accurate representation of the system, *for the particular objectives of the study*. If a model is "valid," then it can be used to make decisions about the system similar to those that would be made if it *were* feasible and cost-effective to experiment with the system itself.

Credibility is when a simulation model and its results are accepted as "correct" by the decision-maker (or manager) and other key project personnel. Validity does not imply credibility and vice versa. For example, a valid or technically correct model might not be used in the decision-making process if the model's key assumptions are not un-

Random Number and Random Variate Generation

Example Ad Hoc Simulation

Spinner

<https://www.google.com/search?q=spinner>
<https://www.visnos.com/demos/random-spinners>

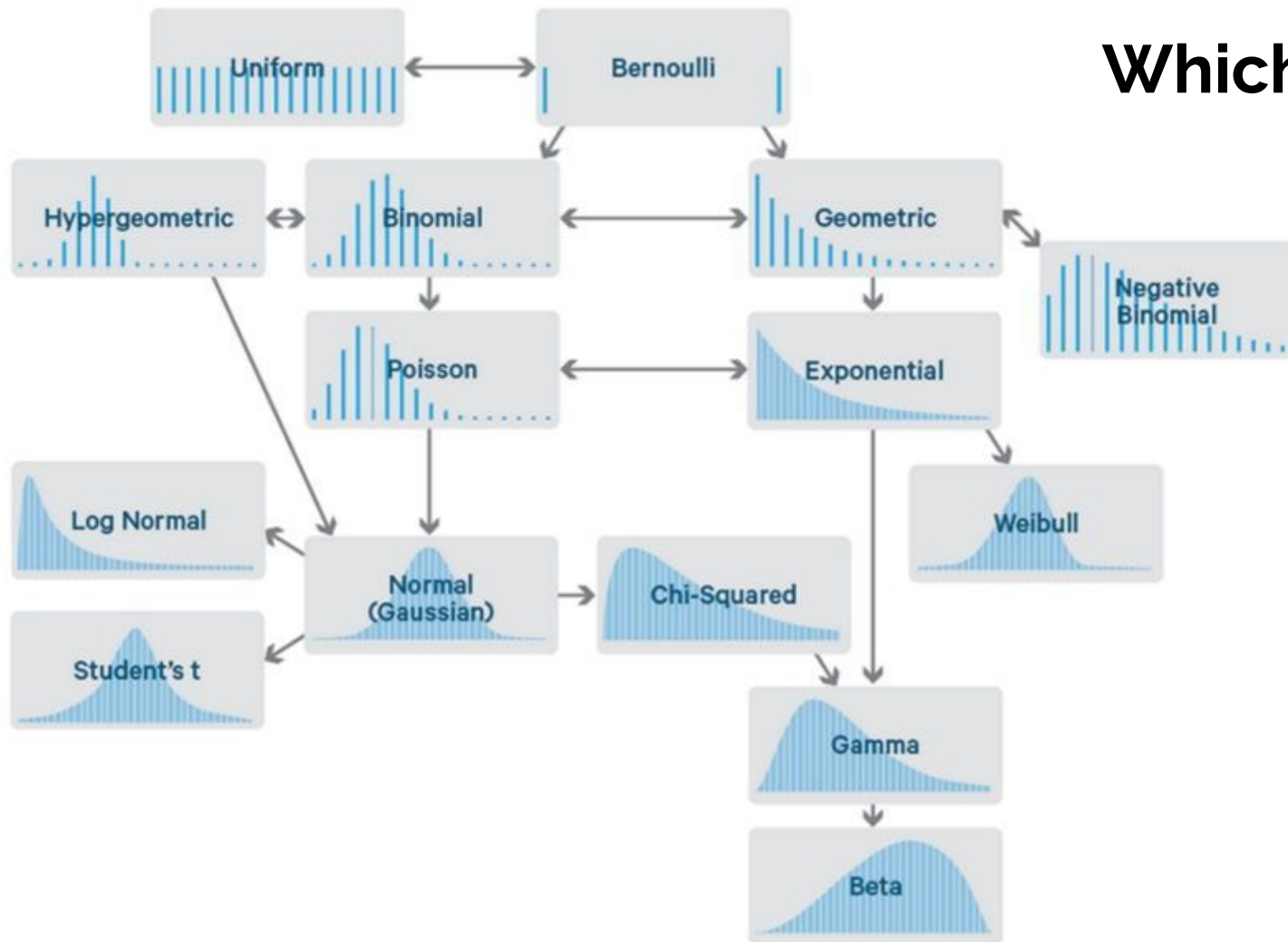


Die

<https://www.google.com/search?q=Roll%20a%20die>
<https://www.online-stopwatch.com/online-dice/>



Which PDF?



Generate exponentially distributed random variables

$$R_i = U(0, 1)$$



$$X_i = -\frac{1}{\lambda} \ln(1 - R_i)$$

- Where X_i is the i th random variate generated [e.g., the time between the arrival of the i th and the $(i+1)$ st entities].
- Suppose that $\lambda = \frac{1}{10}$ arrival per minute.
- If $R_i = 0.3067$, then $X_i = 3.66$ minutes.

—

Input Data

Input Data



The techniques used may vary depending on:

- The amount of available data
- Whether the data are “hard” or someone’s best guess
- Whether each variable is independent of other input random variables, or related in some way to other outputs

In the case of a variable that is independent of other variables, the choices are as follows:

- Assume that the variable is deterministic
 - **Fit a probability distribution to the data**
 - Use the empirical distribution of the data
1. Hypothesize a candidate distribution
[2. Estimate the parameters of the hypothesized distribution](#)
3. Perform a goodness-of-fit test such as the chi-squared test

Input Data

The techniques used may vary depending on:

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- **Use the empirical distribution of the data**

Interval (hours)	Frequency of Occurrence
$0 < x \leq 1.0$	27
$1.0 < x \leq 2.0$	13
$2.0 < x \leq 3.0$	31
$3.0 < x \leq 4.0$	18
$4.0 < x \leq 8.0$	11

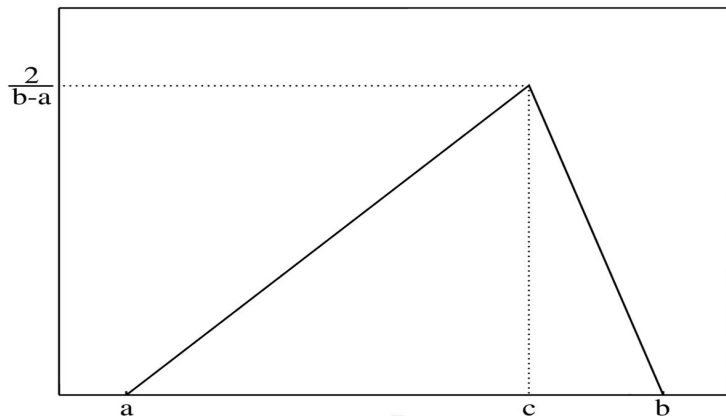
Input Data

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- The amount of available data
- Whether the data are “hard” or someone’s best guess
- Whether each variable is independent of other input random variables, or related in some way to other outputs

In the case of a variable that is independent of other variables, the choices are as follows:

- Assume that the variable is deterministic
- Fit a probability distribution to the data
- Use the empirical distribution of the data
- **When no data are available**



Verification and Validation

Verification and validation



- **Verification**: a determination of whether the computer implementation of the conceptual model is correct. Does the operational model represent the conceptual model?
- **Validation**: a determination of whether the conceptual model can be substituted for the real system for the purposes of experimentation.

The checking processes is iterative.

If there are discrepancies among the operational and conceptual models and the real-world system, the relevant operational model must be examined for errors, or the conceptual model must be modified to represent the real-world system better (with the subsequent changes in the operational model).

Verification



1. Follow the principles of structured programming.
2. Make the operational model as self-documenting as possible.
3. Have the computer code checked by more than one person.
4. Check to see that the values of the input data are being used appropriately.
5. For a variety of input-data values, ensure that the outputs are reasonable.
6. Use an interactive run controller (IRC) or debugger to check that the program operates as intended.
7. Animation is a very useful verification tool.

Validation



1. Face validation.
2. Sensitivity analysis.
3. Extreme-Condition Tests.
4. Validation of conceptual model assumptions.
 - a. Consult with appropriate personnel to determine when peak periods occur.
 - b. Collect interarrival data from these periods.
 - c. Conduct statistical tests to ensure that the assumption of independence is reasonable.
 - d. Estimate the parameter of the assumed exponential distribution.
 - e. Conduct a goodness-of-fit test to ensure that the exponential distribution is reasonable.
5. Consistency checks.
6. Turing tests.
7. Validating Input-Output transformations
8. Validation using historical input data

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Experimentation and Output Analysis

Experimentation and output analysis

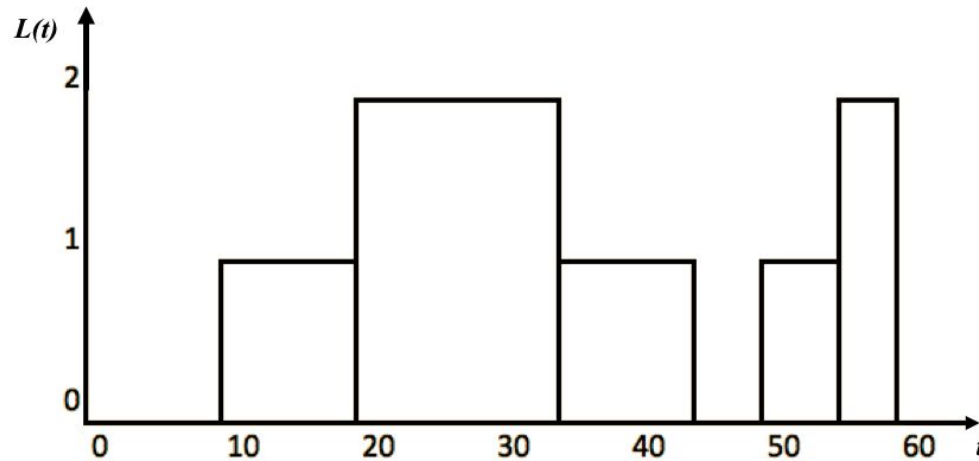


The analysis of simulation output begins with the selection of performance measures.

Performance measures can be time weighted, based on counting of occurrences, or arise from the tabulation of expressions including means, variances, and so on.

Experimentation and output analysis

- An example of a time-weighted statistic is the average number in system over a time period of length T . Figure shows the number in system, $L(t)$, at time t , from $t=0$ to $t=60$.



$$\bar{L} = \frac{(0 \times 10) + (1 \times 10) + (2 \times 15) + (1 \times 10) + (0 \times 5) + (1 \times 6) + (2 \times 4)}{60}$$
$$\bar{L} = 1.07$$

Experimentation and output analysis



- An example of a statistic based on counting of occurrences is the number of acceptable units completed in 24 hours of simulated time.
- A statistic based on the tabulation of expressions is the patent royalties from three different part types, each with a different contribution per unit, for a 24-hour period.

Experimentation and output analysis



Specific questions that must be addressed when conducting output analysis are:

1. What is the appropriate run length of the simulation (unless the system dictates a value)?
2. How do we interpret the simulated results?
3. How do we analyze the differences between different model configurations?

Statistical Confidence and Run Length



A confidence interval for the performance measure being estimated by the simulation model is a basic component of output analysis. A confidence interval is a numerical range that has a probability of $1 - \alpha$ of including the true value of the performance measure, where $1 - \alpha$ is the confidence level for the interval.

For example, let us say that the performance measure of interest is the mean time in the queue, μ , and a $100(1 - \alpha)\%$ confidence interval for μ is desired.

If many replications are performed and the independent confidence intervals on μ are constructed from those replications, approximately $100(1 - \alpha)\%$ of those intervals will contain μ .

Statistical Confidence and Run Length

Example (Confidence intervals). Given the following table, both a 95% ($\alpha=0.05$) and a 99% ($\alpha = 0.01$) two-sided confidence interval are desired. Assuming that the values for X are normally distributed, a $1 - \alpha$ confidence interval for the mean, μ , is given by $(\bar{X} - h, \bar{X} + h)$, where $\bar{X}$ is the sample mean and h is the half-width.

Replication Number, i	Average Time in Queue, X_i
1	63.2
2	69.7
3	67.3
4	64.8
5	72.0

where $t_{n-1, 1-\alpha/2}$ is the upper $1-\alpha/2$ critical value of the t -distribution with $n-1$ degrees of freedom.

$$\bar{X} = \sum_{i=1}^n \frac{X_i}{n}$$

$$h = t_{n-1, 1-\alpha/2} \frac{S}{\sqrt{n}}$$

$$S^2 = \frac{\sum_{i=1}^n X_i^2 - n\bar{X}^2}{n - 1}$$

Experimentation and output analysis



Since a two-sided confidence interval is desired, we use $a/2$ to compute the half-width. Using previous equations, we obtain $\bar{X} = 67.4$ and $S = 3.57$. In addition,

$$t_{4,.975} = 2.78 \text{ (95\% confidence)}$$

$$t_{4,.995} = 4.60 \text{ (99\% confidence)}$$

resulting in

$$h = \begin{cases} 4.44 & \text{(95\% confidence)} \\ 7.34 & \text{(99\% confidence)} \end{cases}$$

The confidence interval is given by $(\bar{X} - h, \bar{X} + h)$. Therefore, the 95% confidence interval is (62.96, 71.84), and the 99% confidence interval is (60.6, 74.74)

Experimentation and output analysis



Three factors that influence the width of the confidence intervals are:

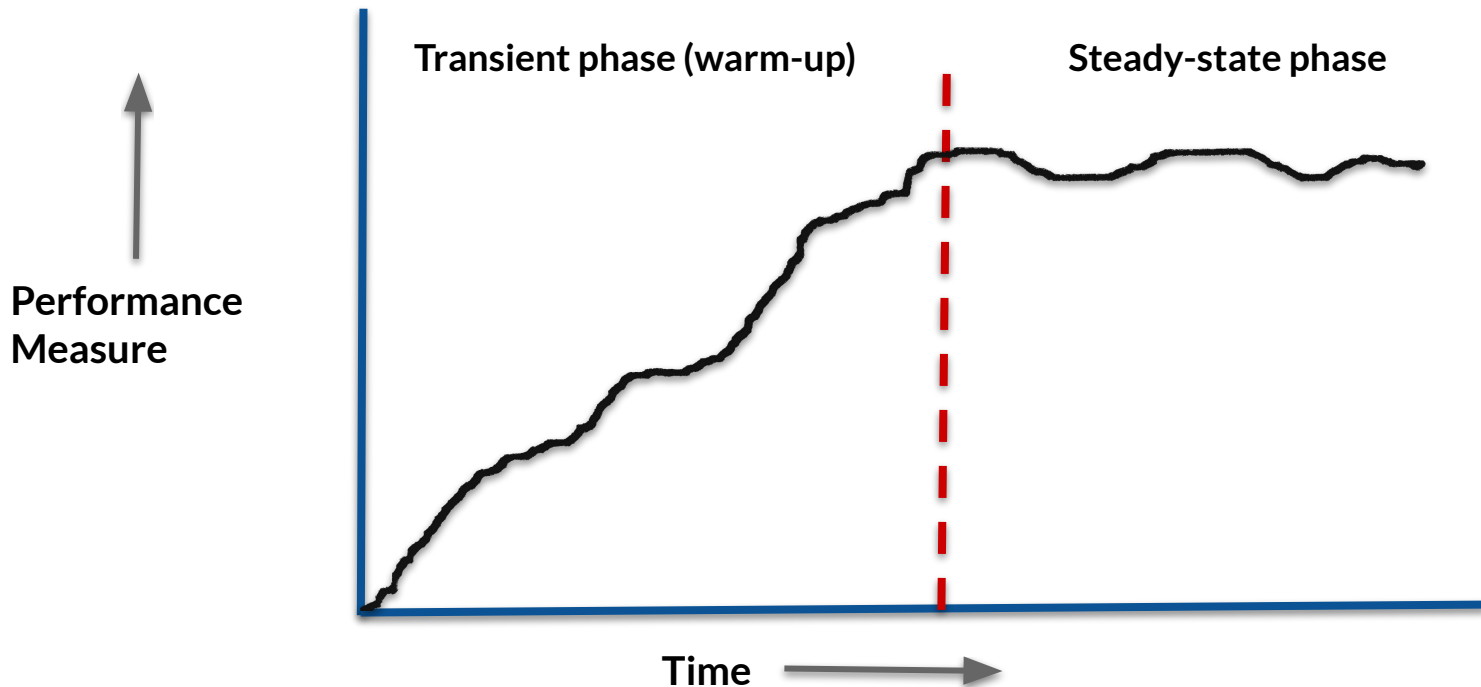
1. Number of replications (n)
2. Level of confidence ($1 - \alpha$)
3. Variation of performance measure (S)

The relationship between these factors and the confidence interval is:

1. As the number of replications increases, the width of the confidence interval decreases.
2. As the level of confidence increases, the width of the interval increases. In other words, a 99% confidence interval is larger than the corresponding 95% confidence interval.
3. As the variation increases, the width of the interval increases.

Terminating Versus Non-terminating Systems

Deletion of initial observations for a nonterminating system.



Reference

[Banks98] Jerry Banks. Handbook of Simulation: Principles, Methodology, Advances, Applications, and Practice. John Wiley & Sons, Inc., 1998

Chap. 1 Principles of Simulation -
[Banks98]

